Haobo Fang

+1 734-972-2668 • fanghb@umich.edu • Homepage

EDUCATION

University of Michigan - Ann Arbor — **Ann Arbor, US**
B.Eng. in Computer Engineering, Cumulative GPA: 3.91 / 4.0 — *Sep. 2022 – Apr. 2026 (expected)*

Shanghai Jiao Tong University — **Shanghai, China**
B.Eng. in Mechanical Engineering, Cumulative GPA: 3.82 / 4.0 — *Sep. 2022 – Aug. 2026 (expected)*

RESEARCH INTERESTS

I am broadly interested in building scalable, adaptable, and reliable robots that interact with humans in daily activities. My research highlights the following perspectives: 1) Origami-Inspired soft robotic arm. 2) Bio-inspired robot. 3) Cutting-edge Control system development. 4) Smarter manufacturing scenarios enhanced by learning.
Research Areas: Soft robotics, Control system, Reinforcement Learning, Manufacturing

EXPERIENCE

Hybrid Dynamic Robotics Lab — **University of Michigan**
Undergraduate Researcher, Advisor: Prof. Xiaonan Huang — *Sep. 2024 - Present*

Origami-Inspired Modular Soft Robotic Arm

- Developed a modular arm based on **Kresling origami actuators**, achieving a predictable linear relationship between contraction and twist.
- Applied the **Piecewise Constant Curvature (PCC) model** for accurate kinematic control with low error (≈2.5%).
- Integrated **CAN-based distributed communication and sensing**, enabling plug-and-play modularity, fault tolerance, and self-perception.

Electromagnetic driven Clip Fish

- Leveraged electromagnetic actuation and bistability to realize the fin's snap-through motion, enabling a compact and agile robotic fish.
- Employed the Discrete Elastic Rod (DER) framework to simulate the tail's motion and optimize its geometric configuration for improved performance.

Data-informed Design and Intelligent Systems Laboratory — **Shanghai Jiao Tong University**
Undergraduate Researcher, Advisor: Prof. Youyi Bi — *Mar. 2023 - May. 2025*

A Zero-shot Robot Tool Manipulation Framework based on Visual Prompt-Enhanced Keypoint Representation and Multi-modal Constraints

- Developed a zero-shot robotic manipulation framework leveraging Vision-Language Models (VLMs) and Visual Foundation Models (VFMs).
- Integrated task decomposition, keypoint localization, multi-modal primitive inference, and real-time error detection to enable generalizable and precise tool use in complex manufacturing scenarios.

Multi-task Robotic Manipulation

- Utilized Reinforcement Learning and Knowledge Transfer to improve the generalization ability and robustness of the multi-task robotic arm in aeroengine assembly scenario.

AWARDS

The Jackson and Muriel Lum Scholarship	University of Michigan, 2025
Dean's Honor List	University of Michigan, 2025
Best Poster Award	ICRA 2025 Workshop, 2025
The Jackson and Muriel Lum Scholarship	University of Michigan, 2024
Huatai Securities Technology Scholarship	Shanghai Jiao Tong University, 2024
Top Student in Morality, Intelligence, and Fitness (top 3%)	Shanghai Jiao Tong University, 2023
C-level Scholarship	Shanghai Jiao Tong University, 2023

PUBLICATIONS

[1] J. Wang, Y. You, X. Zhang, **H. Fang**, J. Wang, and X. Huang, "Origami-Inspired Modular Soft Robotic Arm" *ICRA Workshop on Soft Robotics for Space Applications*, 2025.

[2] J. Wang, Y. You, X. Xhang, **H. Fang**, J. Wang, and X. Huang, 'Origami Inspired Soft Robotic Arm: A Modular Platform for Manipulation" *Extended Abstract, RoboSoft*, 2025.

PATENTS

[1] J. Wang, Y. You, X. Xhang, **H. Fang**, J. Wang, and X. Huang, "Lightweight, Proprioceptive, Origami-Inspired Soft Robotic Arm for High Payload, Low-Cost Reconfigurable Manipulation," *U.S Patent*, *Pending*, 2025.

SERVICE

[1] **Instructional Assistant**, ECE Division, University of Michigan, Fall 2025.
Served as IA for EECS 216: Introduction to Signals and Systems.

[2] **Volunteer**, IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, May 2025.

[3] **Captain**, Eighth "Rongchang Chucai" Plan, Shanghai, Spring 2024 – Spring 2025.
Led team activities and coordinated logistics for the eighth cohort.

[4] **Captain Assistant**, Miyuan Youth Volunteer Group, Shanghai, June 2023 – June 2024.
Supported team operations and participated in 2022 Yunnan volunteer teaching campaign.

COMPETITIONS

China International College Students' Innovation Competition **Shanghai Jiao Tong University**
Mar. 2024 – Oct. 2024

Developed an AI-assisted smart teaching platform to enhance education in rural areas.
Won the 2024 Golden Award in the Shanghai Division.

Liming Cup Mechanical Innovation Competition **Shanghai Jiao Tong University**
Mar. 2023 – Jun. 2023

Developed a multi-task racing car.
Won the 2023 Best Design Award and Second Prize.

SKILLS

Python, C/C++, MATLAB, Rust, PyTorch, Unity, Solidworks, COMSOL Multiphysics